# Collapse, Convergence, and the Scale of the Prime-Gap Imprint on Polymer Dynamics

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A self-avoiding chain whose bending stiffness is programmed by the prime-gap sequence (the prime-driven topological polymer, PDTP) collapses into a state that is measurably less radially sorted than a chain programmed by a random permutation of the same gaps. We settle two questions about this imprint. First, is it a property of the collapsed state or a memory of how the chain was cooled? Two independent relaxation routes at $N = 4 \times 10^4$, eight-fold slower annealing and a $10^6$-step isothermal hold that drives the chain to a melt-density globule ($R_g = 16.81 \pm 0.03$, packing fraction 0.489), converge on the same nonzero gap, $\Delta\rho_\infty = -0.0269 \pm 0.0033$ ($8.1\sigma$); the gap is present at every size from $N = 10^4$ to $8 \times 10^4$ under the converged protocol. The imprint survives collapse. Second, what does the size-proportional correlation length $\xi \propto n_{\mathrm{gap}}$ of the local-scramble surrogate measure? We show that it measures the readout, not the primes. A retention-scaling theorem for the exactly solvable ideal chain predicts $\xi = c\, n_{\mathrm{gap}}$ for any readout carried by chain-scale composition, and surrogates that contain no prime-specific arithmetic (the smooth prime-number-theorem envelope S6′ and the PNT scale family S6″) reproduce the proportional law of the true primes, with $\xi/n_{\mathrm{gap}} = 0.30$, 0.30 and 0.33 for the true, S6′ and S6″ master curves. We report the estimator dependence of $\xi$ in the Langevin ensembles, which is as large as the differences between arms once the arm contrast is small, and replace the single-point half-crossing by a master-curve collapse in $L/n_{\mathrm{gap}}$. The inference from $\xi \propto n_{\mathrm{gap}}$ to extensive arithmetic order in the primes, made in an earlier unpublished version of this work, is withdrawn. What remains is a smaller, robust statement: the fold registers the composition of the small-prime end of the chain, of which the smooth PNT drift supplies a growing fraction ($f_{S6'} = 0.25$, 0.34, 0.57 at $N = 2, 4, 8 \times 10^4$) and the discreteness of the small primes supplies the rest; at the sequence level the prime chain lies below all ten fixed S6′ sequences ($Z_{\mathrm{seq}} = -3.8$).

## I. INTRODUCTION

The folding of a heteropolymer is governed by its monomer sequence [1, 2]. Periodic sequences drive microphase separation [3, 4]; random sequences produce frustrated, glassy landscapes [5, 6]. The prime numbers supply a sequence that is neither: fully deterministic, globally aperiodic, with a logarithmically decaying density and gap correlations that are neither periodic nor memoryless [7–9]. The prime-driven topological polymer (PDTP) [10] maps the local prime gap onto the local bending stiffness of a coarse-grained self-avoiding chain, and reads the collapsed conformation through the Spearman rank correlation $\rho$ between stiffness and radial distance. Against surrogates that preserve the gap multiset exactly, the true sequence is significantly *less* radially sorted; the effect is robust at every size, under deeper annealing, and at the sequence level [10].

Two questions about that imprint were left open by the first version of the model, and an earlier, unpublished Letter answered one of them wrongly. The first question is whether the imprint is a property of the collapsed state at all. Every published ensemble was produced by simulated annealing, which is a kinetic protocol; a chain cooled through a rugged landscape freezes into whatever basin it reaches, and the basin can depend on the sequence in ways that have nothing to do with the thermodynamics of the collapsed state. Section III tests this objection along two independent relaxation routes and finds that the imprint survives, at a renormalised amplitude, in a dense globule.

The second question concerns scale. The local-scramble surrogate S5 destroys order within blocks of $L$ gaps while keeping the blocks in place; the block length at which its retention of the true signal falls through one half defines a correlation length $\xi$. Measured at four sizes, $\xi$ grows in proportion to the number of gaps, $\xi \approx 0.3\, n_{\mathrm{gap}}$. The Letter read this as "extensive arithmetic order" in the primes. Version 4.0 of the PDTP paper [10] proved that this reading does not follow: for the exactly solvable ideal chain, *any* readout carried by chain-scale structure of the stiffness composition gives $\xi \propto n_{\mathrm{gap}}$, whatever the sequence. Sections IV and V make this quantitative. We re-measure $\xi$ at converged depth, report its dependence on the estimator, and show that two surrogates containing no prime-specific arithmetic reproduce the proportional law exactly. Section VI then asks how much of the imprint the smooth prime-number-theorem (PNT) drift accounts for, and corrects the statistics of that decomposition for the fact that the prime arm is a single fixed sequence.

The paper therefore supports one physical conclusion and withdraws one inference. The imprint is real and survives collapse. Its size-proportional correlation length is a property of the readout, not of the primes.

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## II. MODEL AND PROTOCOLS

Beads $i = 0, \ldots, N-1$ are identified with the integers. For a composite bead in the maximal run of composites bounded by primes $p_{\text{left}} < i < p_{\text{right}}$ the local gap is $g_i = p_{\text{right}} - p_{\text{left}}$; prime beads have $g_i = 0$. The gap sets the bending stiffness through a saturated exponential,

$$k_i = \begin{cases} \min[k_{\text{hinge}} e^{\alpha g_i},\, k_{\max}], & i \text{ composite}, \\ k_{\text{hinge}}, & i \text{ prime}, \end{cases} \tag{1}$$

with $k_{\text{hinge}} = 0.25$, $\alpha = 0.15$, $k_{\max} = 500$. Bonds are harmonic ($k_{\text{bond}} = 45$, $b_0 = 1$), bending is a cosine potential weighted by $k_i$ [11, 12], self-avoidance is Weeks–Chandler–Andersen [13] with the pairwise force capped at 30, and a weak radial term ($k_{\text{conf}} = 0.02$) gathers the globule. Chains are propagated by overdamped Langevin dynamics [14] in reduced units ($\sigma = \varepsilon = k_B = \mu = 1$, $\Delta t = 0.008$, per-bead displacement cap 0.25) under geometric annealing from $k_B T = 0.50$ to 0.003 [15]. The full force field, parameter table and calibration are given in Ref. [10]; nothing is changed here.

*Surrogates.* All surrogates preserve the exact gap multiset. S1 randomises the order of the gaps. S5($L$) partitions the ordered gap sequence into contiguous blocks of $L$ gaps, keeps the blocks in place and randomises the gaps within each block; it interpolates from the true chain at $L = 1$ to S1 at $L \geq n_{\text{gap}} = \pi(N) - 1$. S6′ rank-maps the gaps onto independent exponential keys of mean $\ln p_k^{\text{smooth}}$, $p_k^{\text{smooth}} = k(\ln k + \ln\ln k)$: it reproduces the smooth PNT envelope $\langle g \rangle \sim \ln i$ and erases every fine correlation. S6″ rank-maps the gaps onto keys $\ln p_k \cdot g'_J$ with $g'_J$ resampled from the global multiset of $g/\ln p$: a pure PNT scale family [10].

*Observables.* $\rho$ is the Spearman rank correlation between $k_i$ and the radial distance from the centre of mass; $\Delta R$ is the mean radial distance of the stiffest 20% of beads minus that of the most flexible 20%; $R_g$ is the radius of gyration. The retention of a surrogate S is

$$f_{\text{S}} = \frac{\rho_{\text{S1}} - \rho_{\text{S}}}{\rho_{\text{S1}} - \rho_{\text{true}}}, \tag{2}$$

unity for a true-like conformation and zero for a shuffle-like one. Unless stated otherwise, values are ensemble means ± s.e.m. over $n = 40$ trajectories per arm, $p$-values are two-sided Welch tests and $d$ is Cohen's $d$; Sec. VI supplements these with sequence-level statistics.

*Protocols.* Three depths occur. The *production* protocol is 5000, 5000, 8000 and $10^4$ annealing steps for $N = 1, 2, 4, 8 \times 10^4$ (the depth of the Letter's ensembles). The *converged* protocol is 32 000 steps at every size, the four-fold deeper anneal at which the ordering gap at $N = 4 \times 10^4$ has saturated (Sec. III). The *isothermal hold* is $10^6$ further steps at $k_B T = 0.003$ ($\Delta t = 0.004$, displacement cap removed) applied to annealed conformations at $N = 4 \times 10^4$.

## III. CONVERGENCE: THE IMPRINT SURVIVES COLLAPSE

### A. Where the production ensemble sits on the collapse pathway

Held at the final annealing temperature, a production conformation does not sit still. Its energy falls by 60–75% and converges by $\sim 10^5$ steps to a value reached by all ten trajectories, in both arms, to within 0.3%. The rank correlation rises in both arms and plateaus on the same schedule; fitting $\rho(t) = \rho_\infty - Ae^{-t/\tau}$ gives an identical relaxation time in the two arms, $\tau_{\text{relax}} = 1.68 \times 10^4$ steps $= 67\,\tau_{\text{LJ}}$, with $\rho_\infty = +0.1635$ (true) and $+0.1935$ (S1). The production anneal at this size spans $64\,\tau_{\text{LJ}}$: the published ensembles were measured after $0.95\,\tau_{\text{relax}}$, the hold endpoint after $60\,\tau_{\text{relax}}$.

The relaxed state is a compact, uniform, near-spherical globule (Table I, Fig. 3a): $R_g = 16.81 \pm 0.03$, flat interior number density $\simeq 1.0\,\sigma^{-3}$, sharp interface at $r \simeq 21\,\sigma$, equivalent-sphere packing fraction $\phi = 0.489 \pm 0.002$, asphericity below 0.06, identical in the two arms. The ten independent endpoints agree in $R_g$ to two parts in a thousand. The representative production conformation of the first PDTP paper, by contrast, has $R_g \approx 111$: a partially collapsed coil at $\sim 10^{-3}$ of globule density. The earlier work measured the imprint on the collapse *pathway*, not at its end.

### B. Protocol independence: the gap renormalises, then saturates

Slowing the anneal by factors of 2, 4 and 8 ($n = 40$ per arm per protocol) moves the ordering gap and then stops moving (Table II, Fig. 1). Between 1× and 4× the gap shrinks by 42%, from $-0.0457$ to $-0.0263$; between 4× and 8× it does not shrink at all ($-0.0263$ versus $-0.0262$, constant-gap $\chi^2 = 0.00/1$). A saturating fit $\Delta\rho = \Delta\rho_\infty + Ae^{-x/\tau}$ in $x = \log_2(\text{protocol})$ gives $\Delta\rho_\infty = -0.0254 \pm 0.0061$. The gap is significant at every protocol (Welch $p \leq 6.4 \times 10^{-6}$), and its effect size is stable across the converged protocols ($d = -1.15, -1.10, -1.16$).

TABLE I. Structure of the relaxed globule at the end of the $10^6$-step hold ($N = 4 \times 10^4$, five endpoints per arm, mean ± s.d.). $R_{\text{eff}} = R_g\sqrt{5/3}$, $\phi$ is the packing fraction of $\sigma$-diameter beads in the equivalent sphere, $z_{\text{nb}}$ the mean number of non-bonded neighbours within $2^{1/6}\sigma$.

| arm | $R_g$ | $R_{\text{eff}}$ | $\phi$ | $z_{\text{nb}}$ | asphericity |
|---|---|---|---|---|---|
| true | $16.82 \pm 0.02$ | $21.71 \pm 0.03$ | $0.489 \pm 0.002$ | 3.39 | $0.045 \pm 0.011$ |
| S1 | $16.80 \pm 0.03$ | $21.68 \pm 0.04$ | $0.490 \pm 0.003$ | 3.40 | $0.041 \pm 0.009$ |

TABLE II. Ordering gap versus annealing length at $N = 4 \times 10^4$, $n = 40$ per arm per protocol; s.e.m. in parentheses on the last digits. The 1× row is the production ensemble.

| steps | $\rho_{\mathrm{true}}$ | $\rho_{\mathrm{S1}}$ | $\Delta\rho$ | $d$ |
|---|---|---|---|---|
| 8 000 (1×) | +0.0942(65) | +0.1399(34) | −0.0457(73) | −1.39 |
| 16 000 (2×) | +0.1506(56) | +0.1822(24) | −0.0315(61) | −1.15 |
| 32 000 (4×) | +0.1659(46) | +0.1922(27) | −0.0263(53) | −1.10 |
| 64 000 (8×) | +0.1563(36) | +0.1825(35) | −0.0262(51) | −1.16 |

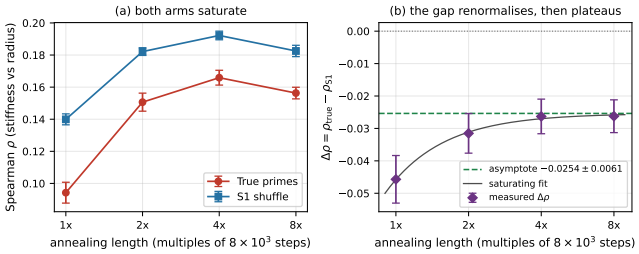



FIG. 1. Protocol independence at $N = 4\times10^4$. (a) Both arms converge: $\rho$ rises from the production protocol and saturates by 4×. (b) The ordering gap renormalises downward by 42% between 1× and 4× and is then constant; green band: the asymptote of the saturating fit.

### C. Isothermal metastability

Over the hold, $\rho$ rises by +0.0147 (true) and +0.0090 (S1), both significant against zero at $n = 5$: the production measurement is taken mid-relaxation. But the drift is the approach to a plateau reached well inside the hold (Fig. 2). Beyond $10^5$ steps ($6\,\tau_{\mathrm{relax}}$) the energy is constant to 0.2%, the ensemble-mean $\rho$ moves by less than 0.004 in either arm, and the gap settles at

$$\Delta\rho\big|_{\mathrm{hold}} = -0.0299 \pm 0.0077, \tag{3}$$

flat to 0.9% of its own value over the final 90% of the hold ($p = 0.0067$, $d = -2.46$ at $n = 5$). Relaxation homogenises the ensemble (the between-run s.d. of $\rho$ falls threefold) while preserving trajectory identity (Pearson $r = 0.91$ between initial and final $\rho$): a near-uniform shift of the whole ensemble, not a re-randomisation.

### D. The converged amplitude

The two routes are mechanistically distinct (slower cooling versus isothermal ageing; capped versus uncapped propagator; $n = 40$ versus $n = 5$) and they agree ($z = 0.40$, $p = 0.69$). Combining the three converged estimates,

$$\Delta\rho_\infty = -0.0269 \pm 0.0033 \qquad (8.1\sigma), \qquad \chi^2 = 0.18/2. \tag{4}$$

This is 59% of the production value: two-fifths of the published separation was incomplete relaxation. Under the converged (32 000-step) protocol the gap is present at every size (Table III): $\Delta\rho = -0.026, -0.039, -0.038, -0.041$ for $N = 1, 2, 4, 8 \times 10^4$, all at $p \le 3 \times 10^{-4}$. The core–shell amplitude $\Delta R$, by contrast, does not survive collapse: it falls from $\simeq 14$ to $2.6\,\sigma$ as the globule compacts and the arm difference ($-0.24\,\sigma$, $p = 0.14$) is unresolved (Fig. 3b). In a globule whose density profile is flat out to a sharp interface there is little radial room for the stiffness extremes to separate; the rank estimator, which uses the whole distribution, retains its resolving power.

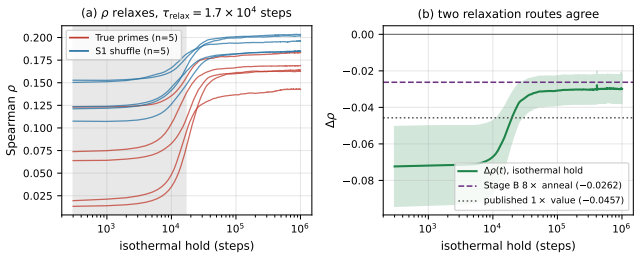



FIG. 2. Isothermal hold at $k_BT = 0.003$, $N = 4 \times 10^4$, $n = 5$ per arm, $10^6$ steps. (a) Individual $\rho$ traces; both arms relax with $\tau_{\mathrm{relax}} = 1.68 \times 10^4$ steps (shaded). (b) The ordering gap shrinks during the relaxation and is then flat; its plateau coincides with the independently obtained 8×-anneal value (dashed).

TABLE III. Ordering gap at the converged protocol (32 000 steps, $n = 40$ per arm) at four sizes, from the same ensembles as the S5 ladders of Sec. IV.

| $N$ | $\rho_{\mathrm{true}}$ | $\rho_{\mathrm{S1}}$ | $\Delta\rho$ | $p$ |
|---|---|---|---|---|
| $1 \times 10^4$ | +0.0774 | +0.1037 | −0.0263 | $2.9 \times 10^{-4}$ |
| $2 \times 10^4$ | +0.1074 | +0.1460 | −0.0386 | $7.3 \times 10^{-9}$ |
| $4 \times 10^4$ | +0.1602 | +0.1982 | −0.0380 | $8.6 \times 10^{-10}$ |
| $8 \times 10^4$ | +0.1978 | +0.2388 | −0.0411 | $8.2 \times 10^{-17}$ |

The kinetic-artefact objection is thus answered in the direction the model needed. A frozen memory of cooling would decay monotonically as cooling slows and would not survive a five-fold contraction of the chain and a 70% drop in energy. This gap decays by 42% and then stops, and survives both. Neither route is equilibrium sampling, which is out of reach at these sizes (the production force field is non-conservative through its force and displacement caps, and a replica-exchange ladder at $N = 4\times10^4$ would need $\sim 10^3$ replicas); what is established is protocol independence of a converged, compacted state along two independent routes.

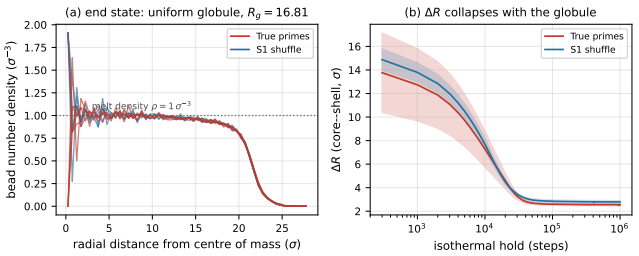



FIG. 3. The relaxed state. (a) Radial bead-density profiles of the ten endpoint configurations: a uniform interior at melt density with a sharp interface at $r \simeq 21\,\sigma$, identical in both arms. (b) The core–shell amplitude $\Delta R$ collapses with the globule and no longer separates the arms.

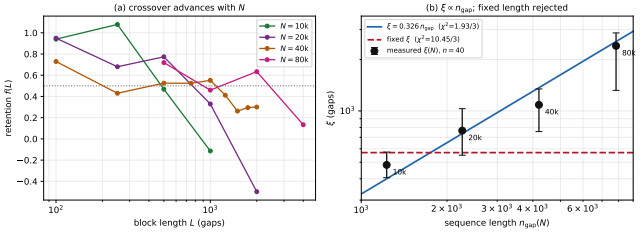



FIG. 4. Production-protocol measurement. (a) Retention $f(L)$ of the S5 local scramble at four sizes, $n = 40$ per point (mean ± s.e.m.); the half-retention crossover (dotted) advances toward larger block length as the sequence grows. (b) $\xi$ against $n_{\rm gap}$ with 68% bootstrap intervals; solid: proportional law; dashed: best fixed length. This apparent growth was interpreted in earlier work as arithmetic order; Fig. 5 and Sec. V show it is reproduced by non-arithmetic surrogates.

## IV. THE CORRELATION LENGTH AND ITS ESTIMATOR

### A. Definition and the production measurement

The S5 local scramble measures the scale over which the ordering signal survives. Its retention $f(L)$ [Eq. (2)] falls from unity at small $L$ to zero at $L \geq n_{\rm gap}$, and the block length at which it falls through one half defines $\xi$. Operationally, in the Letter and here, $\xi$ is the last downward crossing of $f(L)$ through $1/2$, located by linear interpolation in $\log L$; its uncertainty is a 68% percentile interval from 8000 bootstrap resamples of the per-trajectory $\rho$ values in every arm, with the crossing construction repeated on each resample.

At the production protocol (Table IV, Fig. 4) the correlation length grows with the sequence: a 6.4-fold increase in $n_{\rm gap}$ produces a 5.0-fold increase in $\xi$. A single-parameter proportional law fits the four points ($\chi^2 = 1.93/3$) and a fixed length is rejected ($\chi^2 = 10.45/3$, $p = 0.015$). The ratio $\xi/n_{\rm gap}$ lies between 0.26 and 0.39.

### B. The converged measurement and the estimator problem

Re-measured at the converged protocol with the same estimator (Table IV, columns LC), $\xi$ remains macroscopic and a fixed length remains rejected ($\chi^2 = 25/3$), but the ratios change: 0.34, 0.30, 0.42 at $N = 1, 2, 4\times10^4$, and at $N = 8 \times 10^4$ the retention stays close to one half from the smallest block ($L = 50$, $f = 0.38$) to $L = 2299$ ($f = 0.47$) before falling, so that no half-crossing can be assigned. The dependence on protocol at fixed $N$ is not monotone: at $N = 10^4$, $\xi = 350$, 482 and 415 gaps after 2000, 5000 and 32 000 steps; at $N = 2 \times 10^4$ deeper annealing lowers $\xi$, at $4\times10^4$ it raises it. The Letter's argument that the protocol bias is "conservative" because $\xi$ increases with depth is therefore unsupported and is withdrawn.

A second estimator, the half-point of a logistic fit to $f(\ln L)$ with free amplitude, width and centre (the estimator of the converged-ladder driver, Appendix B), gives 0.30, 0.52, 0.52, 0.50 on the same data. The two estimators agree at $N = 10^4$ and disagree by up to a factor two elsewhere; neither is wrong, because the retention curves at converged depth are not monotone, and a single half-point of a non-monotone curve is not well defined.

The problem is sharpest where the arm contrast is small. Applying the S5 ladder to S6′ (Sec. V), the denominator of Eq. (2) is $\rho_{\rm S1} - \rho_{\rm S6'} = 0.010$–$0.023$ against a per-point s.e.m. of $\rho_{\rm S5}$ of $\simeq 0.005$: $f(L)$ then carries errors of 0.3–0.5 per point, takes values up to 1.8, and crosses one half several times. For these ladders the LC estimator gives $\xi/n_{\rm gap} = 0.43$, 0.61, 0.58 and the LG estimator 0.15, 0.15, 0.27. A Langevin measurement of $\xi$ on a weak arm is therefore not a number but a range, and we report it as such.

We adopt two remedies. First, within the Langevin data we replace the single-point crossing by the *master-curve collapse*: $f$ is plotted against $L/n_{\rm gap}$ for all sizes, and the statement tested is whether the curves coincide (Sec. V, Fig. 5c). Second, the decisive comparison between arms is made in the exact ideal-chain readout, where there is no thermal noise and the only variance is that of the surrogate construction.

## V. THE SCALE IS A PROPERTY OF THE READOUT

### A. The retention-scaling theorem

Reference [10] solves the ideal (phantom) chain with the quenched PDTP bending field exactly: the thermally averaged radial profile $P_i = \langle|\mathbf{r}_i - \mathbf{r}_{\rm cm}|^2\rangle$ of any sequence follows from the tangent-product formula $\langle \mathbf{t}_a \cdot \mathbf{t}_b \rangle = \prod_{k=a+1}^{b} \mathcal{L}(\beta k_k)$ in $O(N)$ operations. For a homogeneous chain $P_i$ reduces to the classical end-to-middle U-shaped profile; the ends lie peripherally as a positional fact. The ideal readout $\rho_{\rm id} = \mathrm{Spearman}(k_i, P_i)$ isolates the posi-

TABLE IV. The S5 half-retention length at the production protocol ($5000/5000/8000/10^4$ steps for $N = 1, 2, 4, 8 \times 10^4$, $n = 40$ per arm) with 68% bootstrap intervals, and at the converged protocol (32 000 steps at every size) with two estimators. LC: last downward crossing of $f(L)$ through 1/2; LG: half-point of a three-parameter logistic fit to $f(\ln L)$ (Appendix B). At $N = 8\times10^4$ under the converged protocol the retention stays near one half across the whole ladder ($f = 0.38$–$0.57$ for $L = 50$–$2299$) and the LC estimate is not determined (its bootstrap interval spans $[386, 2490]$).

| | | production | converged | |
|---|---|---|---|---|
| $N$ | $n_{\mathrm{gap}}$ | $\xi$ (LC) | $\xi$ (LC) | $\xi$ (LG) |
| $1 \times 10^4$ | 1228 | 482 [406, 574] | 415 [362, 464] | 364 |
| $2 \times 10^4$ | 2261 | 766 [550, 1031] | 668 [249, 902] | 1176 |
| $4 \times 10^4$ | 4202 | 1086 [757, 1346] | 1774 [1526, 2060] | 2182 |
| $8 \times 10^4$ | 7836 | 2407 [1320, 2869] | undetermined | 3940 |
| $\xi/n_{\mathrm{gap}}$ | | 0.39, 0.34, 0.26, 0.31 | 0.34, 0.30, 0.42, – | 0.30, 0.52, 0.52, 0.50 |

tional channel: how the composition of the sequence is distributed relative to the chain ends.

Let $m_i = \phi(k_i) - \bar{\phi}$ be the composition field ($\phi$ the rank transform, identical across multiset-preserving arms). A readout is *chain-scale* if its arm contrast depends on the sequence through projections $\langle m, F\rangle = \sum_i m_i F(i/N)$ on functions of bounded variation. Theorem 1 of Ref. [10] states that under S5 with blocks of $L$ gaps,

$$\left|\langle m - \bar{m}_L, F\rangle\right| \le \frac{L\bar{g}}{N}\,\mathrm{TV}(F)\,\|m\|_\infty, \tag{5}$$

with $\bar{m}_L$ the block average, so that the retention depends on $L/n_{\mathrm{gap}}$ alone (up to the slow variation of the mean gap $\bar{g}$) and its half-point is $\xi = c_F\, n_{\mathrm{gap}}$. The constant $c_F$ is fixed by the shape of $F$ and the chain-scale content of $m$; it is not an arithmetic quantity. A monotone PNT drift, which spans the whole sequence by construction, is a chain-scale composition profile and therefore suffices. The theorem also permits $\ln n_{\mathrm{gap}}$ corrections through $\bar{g}$, which is why the Letter's distinction between a strictly proportional and a log-corrected law has no content and is not pursued here.

### B. Surrogates without arithmetic reproduce the proportional law

The theorem predicts that S6′ and S6″, which carry the PNT envelope and nothing prime-specific, give the same $\xi/n_{\mathrm{gap}}$ as the primes. We test this in the exact readout at four sizes (`ideal_chain_exact.py`: 40 realisations of every surrogate per block length, 2000 bootstrap resamples; Table V, Fig. 5a,b). Every arm has a negative contrast against S1 ($z = -1.4$ to $-3.3$) and a macroscopic, size-proportional $\xi$. The size-averaged master curves have half-points

$$\frac{\xi}{n_{\mathrm{gap}}} = 0.30\ (\mathrm{true}),\quad 0.30\ (\mathrm{S6}'),\quad 0.33\ (\mathrm{S6}''), \tag{6}$$

and the r.m.s. spread of $f$ across the four sizes on a common $L/n_{\mathrm{gap}}$ grid is 0.08–0.12 for all three arms. The per-size values scatter (0.21–0.40), and the S6′ half-points at $N = 1$ and $2 \times 10^4$ (0.23, 0.21) lie below the true value. Two effects, both finite-size, account for this. First, the S6′ contrast against S1 is smallest at small $N$ ($z = -1.4$), so the denominator of Eq. (2) is small and the single-size half-point inherits the full construction variance of the ensemble, exactly as in the Langevin ladders. Second, the bound of Eq. (5) is not uniform in $N$: its right-hand side carries the mean gap $\bar{g}$ and the total variation of the readout profile, and at $N = 10^4$ the small-prime region, where $\ln p_i$ rises in discrete steps and the gap law is far from a scale family, is a larger fraction of the chain than at $8 \times 10^4$. The constant $c_F$ therefore approaches its asymptotic value only as the chain-end composition becomes a vanishing fraction of the backbone, and S6′, which replaces the discrete small-prime gaps by a smooth envelope, converges from below. Once the curves are normalised to $L/n_{\mathrm{gap}}$ and averaged over sizes these effects cancel: the master-curve half-points of the true primes and of S6′ coincide (0.30 versus 0.30), and the proportional law and its coefficient are the same for the true primes and for the two surrogates that contain no prime-specific arithmetic. This is the result the theorem requires and the result the Letter's inference cannot survive: the growth of $\xi$ with $n_{\mathrm{gap}}$, and its coefficient, are produced by the smooth PNT envelope read through the end-to-middle profile.

In the Langevin ensembles the same collapse holds for the true primes within their errors (Fig. 5c): the converged curves at $N = 1$, 2 and $4 \times 10^4$ lie on one band in $L/n_{\mathrm{gap}}$, and the $8\times10^4$ curve lies below it at small blocks, where a partial loss of retention already at $L = 50$ is not captured by the ideal readout. We return to this point, which we read as a finite-depth kinetic effect, in Sec. VII. The Langevin S6′ ladders, run at $N = 2, 4, 8 \times 10^4$ under the converged protocol, also show a retention that moves toward larger $L$ as the sequence grows, but their small contrast makes the curves too noisy to locate a half-point reliably; we therefore rest the quantitative statement on the exact readout, where the point is settled without noise.

TABLE V. Exact ideal-chain readout: half-retention length of the S5 scramble applied to the true primes, to S6′ and to S6″ (40 realisations per point, 68% bootstrap intervals over the surrogate ensemble; the true arm is one fixed sequence). $z$ is the contrast of the top arm against the S1 ensemble in units of the S1 s.d. Last row: half-point of the size-averaged master curve in $L/n_{\mathrm{gap}}$.

| | true | | S6′ | | S6″ | |
|---|---|---|---|---|---|---|
| $N$ | $z$ | $\xi/n_{\mathrm{gap}}$ | $z$ | $\xi/n_{\mathrm{gap}}$ | $z$ | $\xi/n_{\mathrm{gap}}$ |
| $1\times10^4$ | $-2.4$ | 0.33 [0.29, 0.40] | $-1.4$ | 0.23 [0.20, 0.30] | $-1.7$ | 0.40 [0.34, 0.43] |
| $2\times10^4$ | $-1.6$ | 0.29 [0.28, 0.31] | $-1.4$ | 0.21 [0.19, 0.37] | $-1.8$ | 0.31 [0.26, 0.36] |
| $4\times10^4$ | $-2.5$ | 0.33 [0.31, 0.35] | $-2.2$ | 0.35 [0.32, 0.39] | $-1.9$ | 0.27 [0.23, 0.31] |
| $8\times10^4$ | $-3.1$ | 0.27 [0.25, 0.28] | $-2.4$ | 0.31 [0.28, 0.34] | $-3.3$ | 0.34 [0.32, 0.37] |
| master curve | | 0.30 | | 0.30 | | 0.33 |

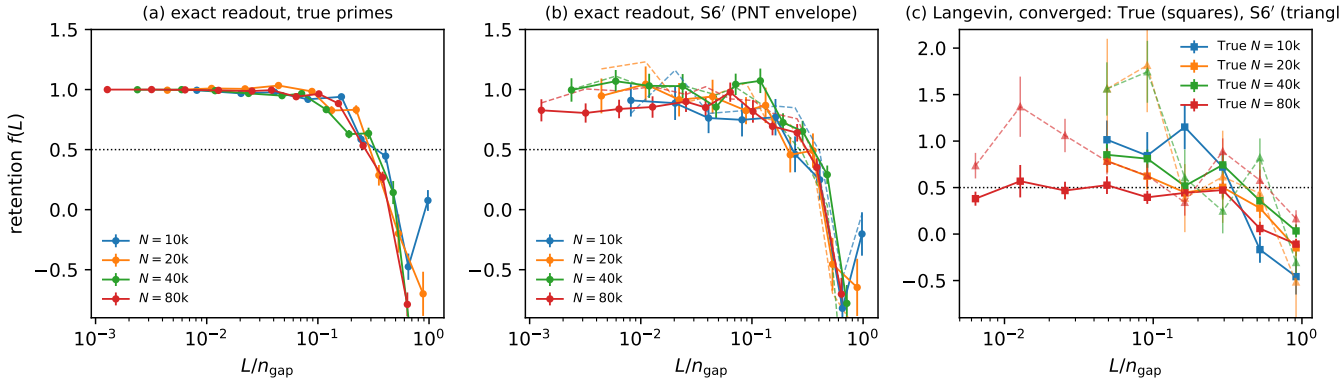



FIG. 5. Retention against $L/n_{\mathrm{gap}}$. (a) Exact ideal-chain readout, true primes at four sizes (error bars: bootstrap s.d. over the S1 and S5 ensembles). (b) Same for S6′ (symbols) and S6″ (dashed): the surrogates without prime-specific arithmetic collapse onto the same master curve as the primes. (c) Langevin ensembles at the converged protocol: true (squares) and S6′ (triangles). The true curves collapse to within their errors; the S6′ curves are dominated by the small arm contrast (Sec. IV).

### C. What is withdrawn

The Letter inferred, from $\xi \propto n_{\mathrm{gap}}$, that "the arithmetic order the fold registers is extensive" and that the primes carry "scale-free long-range order" visible to the fold. That inference is withdrawn, together with the fixed coefficient 0.326 (the measured ratio depends on protocol and estimator and lies in the range 0.27–0.37 for the true primes at converged depth and in the exact readout), the claim that the protocol bias is conservative, and the discussion of a logarithmic correction. The Letter's conjecture that the fold registers the hyperuniform, limit-periodic order of the primes [16, 17] is not supported either: Ref. [10] shows that spectrum-matched IAAFT surrogates reproduce most of the effect and that the small-wavenumber hyperuniform exponent of the sequence is not transmitted to the scattering function at all.

## VI. THE PNT SHARE AND THE RESIDUAL

### A. Decomposition at production and at converged depth

Macroscopic order in the gap sequence has two layers: the smooth PNT drift of the mean gap, $\langle g_i\rangle \sim \ln i$, and the deterministic fluctuation riding on it. S6′ carries the first and not the second, so $f_{S6'}$ [Eq. (2)] is the fraction of the ordering effect reproduced by the envelope alone (Table VI). At the production protocol it is 0.32 and 0.44 at $N = 2$ and $4\times10^4$. At the converged protocol it is 0.25, 0.34 and 0.57 at $N = 2$, 4 and $8\times10^4$: the share of the smooth envelope grows with $N$, as a drift that spans the whole chain should, while the residual true−S6′ gap remains significant at every size ($p \le 10^{-4}$) and shrinks in absolute terms from $-0.029$ to $-0.018$.

Two cautions from Ref. [10] apply. S6′ reproduces only 50–80% of the true chain-scale composition (the leading Legendre coefficients of rank$(k_i)$), so $1 - f_{S6'}$ is an upper bound on the non-PNT share. The improved scale family S6″, which reproduces those coefficients to 0.93–1.0 and recovers the full effect in the ideal readout, was

TABLE VI. PNT decomposition. $f_{S6'} = (\rho_{S1} - \rho_{S6'})/(\rho_{S1} - \rho_{true})$, $n = 40$ per arm; $p$ and $d$ refer to the true−S6′ contrast (Welch, fresh S6′ sequence per trajectory). For the converged $N = 2\times10^4$ row the sequence-level design of Sec. VI C (Table VII) gives $Z_{seq} = -3.75$, $p = 4.6 \times 10^{-3}$, and $f_{S6'} = 0.46$ [0.32, 0.59] with fixed S6′ sequences.

| protocol | $N$ | $\rho_{true}$ | $\rho_{S6'}$ | $\rho_{S1}$ | $f_{S6'}$ | true−S6′ ($p$, $d$) |
|---|---|---|---|---|---|---|
| production | $2 \times 10^4$ | +0.0396 | +0.0654 | +0.0777 | 0.32 | $-0.0258$ ($3.8 \times 10^{-2}$, $-0.47$) |
| production | $4 \times 10^4$ | +0.0942 | +0.1197 | +0.1399 | 0.44 | $-0.0254$ ($4.0 \times 10^{-3}$, $-0.66$) |
| converged | $2 \times 10^4$ | +0.1074 | +0.1364 | +0.1460 | 0.25 | $-0.0291$ ($8.9 \times 10^{-6}$, $-1.06$) |
| converged | $4 \times 10^4$ | +0.1602 | +0.1851 | +0.1982 | 0.34 | $-0.0250$ ($1.3 \times 10^{-5}$, $-1.04$) |
| converged | $8 \times 10^4$ | +0.1978 | +0.2155 | +0.2388 | 0.57 | $-0.0177$ ($5.2 \times 10^{-5}$, $-0.96$) |

predicted to give $f \approx 1$ in the Langevin model and gave 0.62 [0.38, 0.94] at $N = 2 \times 10^4$ and 0.28 [0.11, 0.45] at $4 \times 10^4$ (production protocol). A smooth model of the composition does not reproduce the effect in the interacting globule.

### B. Where the residual lives

Reference [10] locates the residual directly: restricting the Spearman readout to bead subsets, the true−S1 gap at $N = 4\times10^4$ is $-0.045$ on the whole chain, $-0.018$ with the first 5% of beads excluded and $+0.002$ ($p = 0.89$) with the first 10% excluded, in both production and isothermally relaxed ensembles; the two halves carry opposite signs of the form the positional readout predicts. The true−S6″ residual is likewise concentrated at the start ($-0.027$ overall, $-0.013$ with the first 10% excluded). The fold reads the composition of the small-prime end of the chain, where the PNT drift is steepest and the gaps are short and regular. The smooth envelope accounts for a growing fraction of this as $N$ increases (Table VI); the rest is the discreteness of the small primes, which no $\ln p$-scaled model reproduces.

### C. Sequence-level statistics

The true arm is one fixed sequence, while every S6′ and S1 trajectory above drew a fresh surrogate sequence. Writing $\rho = \mu + s + \varepsilon$ with a sequence effect $s$ of variance $\sigma^2_{seq}$ and thermal noise $\varepsilon$, the variance of $\bar\rho_{true} - \bar\rho_S$ under the null "the primes are a typical member of the class S" contains $\sigma^2_{seq}$, which does not shrink with $n$ and which the Welch statistic omits [10]. For the shuffle class at $N = 10^4$ the omitted term is undetectable next to thermal noise (ANOVA $p = 0.95$; the prime chain lies below all twelve fixed shuffles, $Z_{seq} = -7.0$) [10]. For the S6′ class it has not been measured. We therefore run $K = 10$ fixed S6′ sequences with $m = 5$ trajectories each at $N = 2 \times 10^4$ under the converged protocol (`random_effects_s6prime.py`), and report

$$Z_{seq} = \frac{\bar\rho_{true} - \hat\mu}{\sqrt{\hat\sigma^2_{seq} + s^2_{true}/n_{true} + \mathrm{Var}\,\hat\mu}}, \tag{7}$$

TABLE VII. Sequence-level test of the S6′ class at $N = 2 \times 10^4$, converged protocol (32 000 steps): $K = 10$ fixed S6′ sequences $\times$ $m = 5$ trajectories, against the true chain ($n = 40$) and the fresh-sequence S1 arm ($n = 40$) of Table III.

| | |
|---|---|
| $\hat\mu$ (class mean) | +0.1284 |
| $\hat\sigma_{seq}$, $\hat\sigma_{th}$ | 0.000, 0.027 |
| ANOVA $F$ (9, 40), $p$ | 0.54, 0.84 |
| $\bar\rho_{true} - \hat\mu$ | $-0.0210$ |
| $Z_{seq}$ ($p$, 9 d.o.f.) | $-3.75$ ($4.6 \times 10^{-3}$) |
| rank of true among sequence means | 0/10 |
| naive Welch $p$ (pooled trajectories) | $2.5 \times 10^{-4}$ |
| $f_{S6'}$, sequence-level 68% interval | 0.46 [0.32, 0.59] |

$$\mathrm{Var}\,\hat\mu = \frac{\hat\sigma^2_{seq} + \hat\sigma^2_{th}/m}{K}, \tag{8}$$

with $\hat\sigma^2_{seq} = \max[0, (MS_{between} - MS_{within})/m]$ from the one-way ANOVA, together with the rank of the prime chain among the $K$ sequence means and a sequence-level bootstrap interval on $f_{S6'}$ (sequences resampled, then trajectories within sequences).

The result (Table VII) is that the between-sequence variance of the S6′ class is undetectable next to thermal noise: the ten sequence means lie between +0.108 and +0.139 with a spread fully accounted for by $\hat\sigma_{th}/\sqrt{m} = 0.012$ (ANOVA $F = 0.54$, $p = 0.84$, $\hat\sigma_{seq} = 0$). The random-effects correction therefore does not weaken the contrast: the prime chain (+0.107) lies below all ten fixed S6′ sequences, $Z_{seq} = -3.75$ ($p = 4.6 \times 10^{-3}$ on 9 d.o.f.), against a naive Welch $p = 2.5\times10^{-4}$. The fixed-sequence class mean, $\hat\mu = +0.128$, is 0.008 below the fresh-per-trajectory S6′ arm of Table VI (+0.136; the difference is 1.4 s.e.m. and we quote both), which puts the envelope's share at this size at $f_{S6'} = 0.46$ with a sequence-level 68% interval [0.32, 0.59], bracketing the fresh-sequence value 0.25 and the production value 0.32. The qualitative conclusion is unchanged by the design: at $N = 2 \times 10^4$ the smooth envelope reproduces between a quarter and a half of the effect, and the residual is significant at the sequence level.

## VII. DISCUSSION

The results of Secs. III–VI fit into one picture with the mechanism of Ref. [10]. A collapsing chain places its ends peripherally. A sequence whose stiffness distribution changes steeply near one end is sorted differently from one whose composition is homogeneous along the backbone, and for the primes that change is the small-prime region. The resulting suppression of $\rho$ is modest, a few percent of the radial-rank variance, but it is a property of the collapsed state: it survives eight-fold slower cooling, a $10^6$-step isothermal hold into a melt-density globule, and a threefold homogenisation of the ensemble, and it is present at every size under the converged protocol.

The scale of the imprint is where the earlier version of this work went wrong, and the error is instructive for any polymer probe of sequence order [18]. A readout that is sensitive to position along the chain converts the non-stationary part of a sequence into an apparently macroscopic correlation length that grows with the system. For the primes the non-stationary part is the PNT drift. The S5 scramble, which keeps blocks in place, retains that drift until the blocks become a finite fraction of the chain, so its half-retention length is a fixed fraction of $n_{\rm gap}$ for any sequence with a chain-scale composition profile. The exact readout makes this concrete: surrogates with no arithmetic beyond the envelope give the same master curve and the same coefficient as the primes. Surrogate tests of such probes must therefore control composition as a function of position, not only the marginal, and the correct comparison for a scale claim is between arms on a common $L/n_{\rm gap}$ axis, not a fit of one arm's $\xi(N)$ to a power law.

A methodological lesson accompanies the physical one. The half-retention length is a ratio-of-differences statistic. When the arm contrast in the denominator is small, as it is for S6′ (0.01–0.02) or at large $N$ under deep annealing, the estimate depends on the estimator by a factor of two and its bootstrap interval becomes bimodal. We have reported both estimators and used the exact readout, which has no thermal noise, for the decisive comparison.

*The small-block loss of retention at $N = 8 \times 10^4$.* Under the converged protocol the S5 retention at the largest size is already 0.38 at $L = 50$ gaps (Fig. 5c), whereas the exact readout gives 1.00 there and the three smaller sizes give 0.8–1.0 at comparable $L/n_{\rm gap}$. We read this as a kinetic effect of finite annealing depth rather than a failure of the positional readout, for three reasons. First, the relaxation time of $\rho$ measured by an isothermal hold after the production anneal is flat in $N$ ($\tau_{\rm relax} = 68$, 68, 66 and 58 $\tau_{\rm LJ}$ at $N = 1, 2, 4, 8 \times 10^4$, each within 1.4 of the confinement-mode time $1/\mu k_{\rm conf} = 50\,\tau_{\rm LJ}$), so the 32 000-step protocol is about $2\,\tau_{\rm relax}$ at every size and a Rouse-type $\tau \propto N^2$ argument does not apply to this force field; the ordering gap itself is saturated at every size (Table III). Second, what does change with $N$ at matched steps is the depth of compaction: the equivalent-sphere packing fraction at the end of the 32 000-step anneal is 0.41, 0.36, 0.30 and 0.32 at the four sizes, against 0.49 in the fully relaxed globule, so the two largest chains are measured in a denser, more slowly rearranging but still incompletely collapsed state at 0.4 steps per bead. The S5(50) chains at $8 \times 10^4$ have the same $R_g$ and packing fraction as the true chain (24.5 and 0.316 versus 24.4 and 0.320), so the loss of retention is not a density artefact. Third, S5 at $L = 50$ gaps ($\simeq 550$ beads) leaves the chain-scale composition intact and changes only the local stiffness pattern within each block; in a dense, partially collapsed globule the local rearrangements needed to bring a scrambled block to the same radial placement as the true block are the slow mode that the single-exponential $\tau_{\rm relax}$ of $\rho$ does not resolve (a second mode with $\tau \sim 6 \times 10^5$ steps and small amplitude was seen in the isothermal hold). A mild local kinetic trapping of the scrambled blocks, growing with $N$ because the per-bead depth falls, is therefore the natural reading. It is testable at laptop cost: extend the $N = 8 \times 10^4$ ladder to the calibrated 48 000-step depth, or hold the S5(50) and true endpoints isothermally and ask whether their $\rho$ values converge. Until that is done we treat the $8 \times 10^4$ small-block point as a protocol effect and exclude it from the scale statements, which rest on the exact readout. The sequence-level variance of the S6′ class, the last statistical gap in the decomposition, is measured in Sec. VI C and found to be negligible.

*Limitations.* All results use one stiffness map, one $\alpha$ and one confinement strength. The ensembles are annealed, not equilibrium samples; the convergence established is that of the annealed ensemble along two routes. The isothermal hold has $n = 5$ per arm. The exact theory describes the ideal chain, which isolates the positional channel but lacks the generic stiffness sorting of the collapsed globule; the failure of S6″ in the Langevin model shows that the interacting globule reads more of the small-prime composition than the smooth envelope. We make no claim that prime numbers are physically privileged.

## VIII. CONCLUSION

The prime-gap ordering, as opposed to the gap composition, leaves an imprint on the collapsed PDTP that survives complete relaxation: $\Delta\rho_\infty = -0.0269 \pm 0.0033$ at $N = 4 \times 10^4$ from two independent routes, and a significant gap at every size from $10^4$ to $8 \times 10^4$ under the converged protocol. Its size-proportional correlation length is not evidence of scale-free arithmetic order. The retention-scaling theorem predicts $\xi \propto n_{\rm gap}$ for any chain-scale readout, and surrogates carrying only the smooth PNT envelope reproduce the law and its coefficient ($\xi/n_{\rm gap} = 0.30$, 0.30, 0.33 for the true, S6′ and S6″ master curves). The smooth envelope supplies a growing fraction of the imprint with $N$ ($f_{S6'} = 0.25$ to 0.57);

the remainder is the discreteness of the small primes at the start of the chain, and it survives a sequence-level test against ten fixed S6′ sequences ($Z_{\rm seq} = -3.75$, rank 0/10). The folded polymer is a physical readout of position-dependent composition, and that is what it has been reading all along.

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The manuscript, scripts and data of this paper are archived at doi:10.5281/zenodo.23076433; the repository is https://github.com/Ruqing1963/prime-gap-imprint-collapse-and-scale. The three scripts reuse, without modification, the exact solver and Langevin engine of the PDTP repository, version 4.0 (doi:10.5281/zenodo.22944122; https://github.com/Ruqing1963/prime-driven-topological-polymer).

## Appendix A: Converged retention ladders

Table VIII lists the retention $f(L)$ of the S5 scramble at the converged protocol with its propagated s.e.m., from which the entries of Table IV are computed.

TABLE VIII. S5 retention $f(L)$ at the converged protocol (32 000 steps, $n = 40$ per point except $L = 100$ and 200 at $N = 8 \times 10^4$, $n = 13$ and 28). The propagated s.e.m. per point is 0.14–0.25 at $N = 10^4$, 0.09–0.14 at $2 \times 10^4$, 0.07–0.12 at $4 \times 10^4$ and 0.06–0.17 at $8 \times 10^4$.

| $N = 1 \times 10^4$ | | $2 \times 10^4$ | | $4 \times 10^4$ | | $8 \times 10^4$ | |
|---|---|---|---|---|---|---|---|
| $L$ | $f$ | $L$ | $f$ | $L$ | $f$ | $L$ | $f$ |
| 60 | 1.01 | 111 | 0.79 | 205 | 0.85 | 50 | 0.38 |
| 112 | 0.84 | 206 | 0.62 | 384 | 0.81 | 100 | 0.57 |
| 200 | 1.15 | 369 | 0.45 | 685 | 0.52 | 200 | 0.47 |
| 360 | 0.72 | 663 | 0.50 | 1233 | 0.74 | 383 | 0.53 |
| 641 | −0.17 | 1179 | 0.28 | 2192 | 0.36 | 715 | 0.40 |
| 1121 | −0.46 | 2064 | −0.15 | 3836 | 0.03 | 1277 | 0.44 |
| | | | | | | 2299 | 0.47 |
| | | | | | | 4087 | 0.06 |
| | | | | | | 7153 | −0.11 |

## Appendix B: The two estimators

*Last crossing (LC).* With $f_j = f(L_j)$ on the measured ladder, $\xi$ is the last index $j$ with $f_j \geq \frac{1}{2} > f_{j+1}$, interpolated linearly in $\log L$; bootstrap resamples with no crossing, or a crossing outside $(20, 2 \times 10^4)$, are discarded and the fraction discarded is reported. *Logistic (LG).* $f(\ln L)$ is fitted by $f_0/[1 + e^{(\ln L - \ln \xi)/w}]$ with free $f_0$, $\xi$ and $w$, and $\xi$ is the point where the fit equals $f_0/2$; if the fit fails the first linear crossing is used, and failing that the bootstrap median. With a free amplitude, points with $f > 1$ at small $L$ pull the half-point inward, which is why LG gives shorter $\xi$ than LC on the S6′ ladders. *Master curve.* $f$ is interpolated in $\log(L/n_{\rm gap})$ onto a common grid for every size, the r.m.s. spread across sizes is reported, and the half-point of the size-averaged curve is taken by the LC rule (Table V).

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